

What Goes on in the Courtroom? Audiovisual Analysis of Appellate Oral Arguments

Aniket Kesari and Elliott Ash

Overview



1. Theoretical and Empirical Investigations About the Role of Appellate Oral Arguments
2. Multimodal AI Pipeline for Analyzing Video
3. Computational Narrative of Courtroom Oral Arguments
 - Within-case detection of key moments
 - Across case behaviors
4. Empirical applications
 - Are there differences in questioning patterns directed toward the winners and losers of cases?
 - How do judges and attorneys behave in person v. on zoom?
 - How do judges react to attorney gender?
 - Do judges appointed by Presidents of different political parties behave differently?

Research Question: What Goes on in the Courtroom?

- Primarily a descriptive question about how judges and attorneys interact in the appellate oral argument setting
- Previously difficult to study this question *quantitatively* because of the complexity of audiovisual data, even when it is available
- Takeaways for Today:
 - Computational approaches to automatically creating courtroom narratives
 - Initial findings on how judicial behaviors might differ across modality, gender, and political party
- Future Work:
 - Exploring inferential and predictive questions about whether these behaviors matter for decision-making





Theoretical and Empirical Questions About Oral Arguments

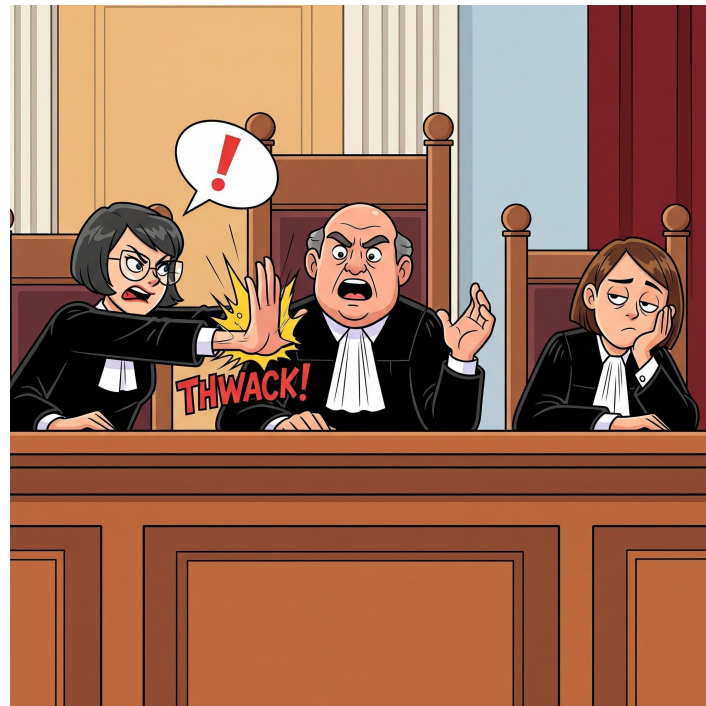
Are Oral Arguments For Information Gathering?



- Oral argument might be about eliciting information that is not available in the brief, or clarifying complex legal questions (Johnson 2001, Johnson et. al. 2006, Wrightsman 2008)
- But not all types of information gathering are the same and can be aimed at different types of questions like clarifying facts, sharpening arguments, etc. (Phillips and Carter 2010ab)
- Sometimes questions might be outright hostile (Epstein 2013)

Are Oral Arguments For Deliberation?

- Alternate view is that there is very little extra information that can be gained in oral argument, especially in an era with ample info in the briefs, amici etc.
- Instead oral argument about giving the judges an opportunity to talk to each other through the attorneys and attempt to persuade their colleagues
- In SCOTUS, evidence of more frequent interruptions, advocacy, and commentary (Jacobi and Sag 2019)



Demographics, Decision-making, and the Workplace

- Political science literature about how demographics like gender and race matter for decision-making
 - Do women judges vote differently in sex discrimination cases (Boyd et al. 2010)
 - Do male judges with daughters vote in more feminist directions (Glynn and Sen 2015)
 - Does the presence of a minority judge influence their colleagues votes (Kastellec 2013)
 - Do audio features predict SCOTUS Justice votes (Dietrich, Enos, and Sen 2019)
 - Do Republican-appointed women and minority judges vote more conservatively than Republican-appointed white males (Copus et al. 2025)?
- Economics literature about courtrooms as workplaces
 - Do “gender slanted” judges treat their female colleagues differently (Ash et. al. 2024)
 - Do judges rationally adjust their behavior when faced with workplace constraints like workload, collegiality etc. (Epstein, Landes, and Posner 2013)



AI Pipeline Overview

Data

- ~15,000 videos of oral arguments at the 9th Circuit from 2010-2025
- Sampled ~500 videos for these results
- Supplement with case disposition data



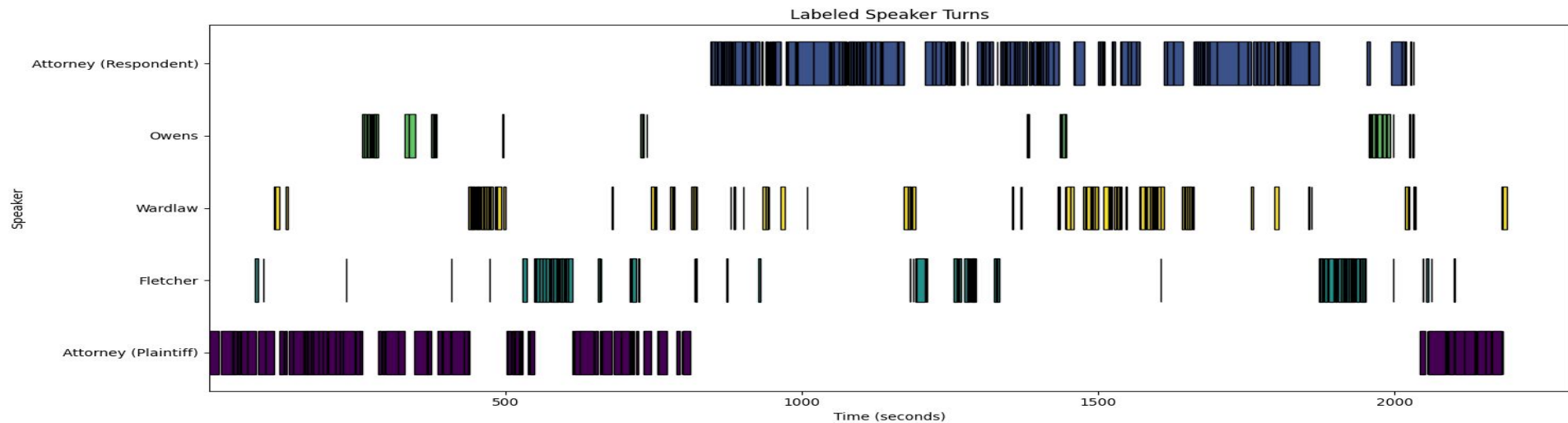
Multimodal AI

- Large Language Models (LLMs) increasingly becoming part of the empirical legal studies landscape (Choi 2024)
- Most recent models are also capable of *multimodal* reasoning and generation (text, audio, and image/video)
- We leverage Gemini 2.5 Pro for its video analysis capabilities



Transforming Video into Data - Transcription

- **Diarization:** Distinguish speakers from one another by using their distinctive audio features
- **Transcription:** Generate a verbatim transcript of what is being said during each “speaker turn”
- **Speaker Identification:** Identify the speaker of each utterance by name



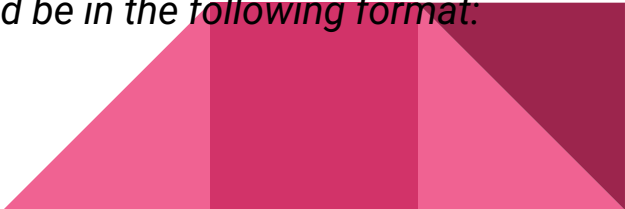
Prompting to Generate a Transcript

Fully transcribe this video of a courtroom oral argument with timestamps and identifying each judge and attorney by name. If it is a zoom proceeding, look to the names in the Zoom boxes and a green outline around the speaker's box to help determine who is speaking. If it is an in-person proceeding, look for a black box specifying the three judges and use these to determine who is speaking when.

Try to group utterances together such that if a speaker is uninterrupted all utterances are associated with the same timestamp. Be attentive to rapid changes in speaker during back-and-forth questioning. For speaker identification, you should use heuristics like:

- 1. If the individual says "Your Honor" they are definitely an attorney and never a judge.*
- 2. The two attorneys should be the most frequent speakers across the whole video.*
- 3. If they are interrupting and asking a question they are probably a judge.*

*Return only the transcript with no other information and each line should be in the following format:
[H:MM:SS] SPEAKER: text*



Transcription Demonstration

[00:00:30] Judge Smith: You don't doubt that this is a, uh, that this is a divisible statute?

[00:00:34] Attorney Vikram K. Badrinath: Your Honor, that's an excellent question.

[00:00:36] Judge Smith: Thank you.

[00:00:37] Attorney Vikram K. Badrinath: It is our position, Your Honor, that, uh, the statute in question is, uh, indivisibly overbroad and therefore it is not a divisible statute.

[00:00:47] Judge Smith: What does it mean to be indivisibly overbroad?

[00:00:50] Attorney Vikram K. Badrinath: Your Honor, I think this, uh, question is illustrated by the Supreme Court's decisions, uh, as reaffirmed, uh, in Descamps, uh, and Moncrieffe, uh, as reaffirming Taylor v. United States, as well as this own, this court's own decision in Rendon v. Holder.



Transforming Video into Data - Structured Output

- Goal is to capture relevant visual and audio data about each actor
- Behavioral psychology and computer science literatures explore what kinds of behaviors might matter
- Body language
 - Posture (leaning forward, slouched, etc.)
 - “Self-touch” (touching elbow, resting head in hand etc., Kronrod 2019)
 - Synchrony (level of behavioral similarity to other actor, Lewis 2010)
- Facial emotional recognition (FER)
 - Anger, happiness, confusion etc. (FER 2013)
- Audio tones
 - Seriousness, skepticism, etc. (Haruvi 2022)



Application: Detecting Key Events Within A Video

1. For each actor, use k-means to cluster their structured behavioral data across a video

- **K-means:** for n observations, sort into k clusters based on minimizing mean distance to centroid

2. Using $k=3$, cluster each actor's behavior and flag as a "key event" when their cluster membership shifts

3. Use a LLM to summarize the key events to construct "computational narratives" of courtroom proceedings

K-means clustering on the digits dataset (PCA-reduced data)
Centroids are marked with white cross

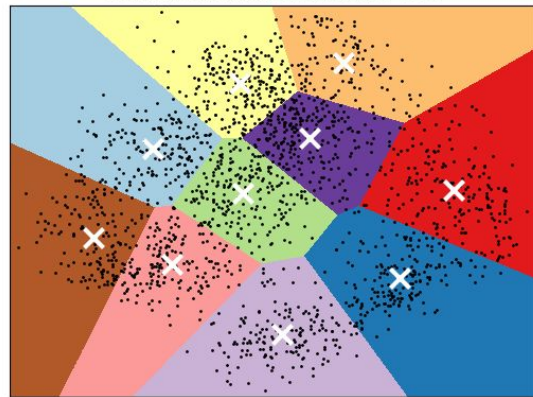


Illustration of k-means algorithm from sklearn library

Key Events Within a Case

Actor Audiovisual Cluster Membership Over Time
Case: Santos Iraheta Martinez v. Garland (Docket: 20-71701)



Key Moments in a Case

Focal Actor	Time	AI Generated Narrative	Video Link
Judge Hurwitz	[01:18:51]	Judge Hurwitz leaned forward and posed a probing question that exposed a potential flaw in the petitioner's argument. His skeptical tone and focused posture heightened engagement.	 The video link column contains three identical stacked images of the James R. Browning U.S. Courthouse Courtroom 2 in San Francisco, California. Each image shows the courtroom with the text 'JAMES R. BROWNING U.S. COURTHOUSE COURTROOM 2' and 'SAN FRANCISCO, CALIFORNIA' overlaid.
Etan Newman	[03:29:43]	Petitioner's counsel presented a confident explanation of flaws in the government's relocation policy. His assured tone and purposeful gestures conveyed conviction; Judge receptive.	
Judge Feinerman	[26:14:14]	Counsel conceded weakness on the IJ's language. Judge Feinerman stepped in to reframe the argument, suggesting a stronger line of reasoning while others listened closely.	

Free-Text Modeling

Overview

- **Data Generation:** AI extraction of visual and audio behavioral cues
 - Series of statements like “The judge strokes his chin” or “The attorney moves his arms”
- **Topic Modeling:** Use BERTopic to group ~200,000 statements into distinct topics
- **Factor Analysis:** Represent each judge as a topic vector
- **Regression Analysis:** OLS to estimate topic share, controlling for judge demographics



Transforming Video into Data - Free Text

You are an expert at identifying non-verbal behavioral cues from both judges and attorneys during courtroom oral arguments. Please watch this video of a courtroom oral argument.

Identify each judge and attorney by name. For attorneys, use heuristics such as when they introduce themselves at the beginning of their opening statements. For judge names and other information, use your knowledge about 9th Circuit judges to identify them and fill in information.

For each actor, please record approximately 100 behaviors specifying things related to body language, facial expressions, vocal tones, and other non-verbal cues. Split these observations between the two attorneys' arguments.

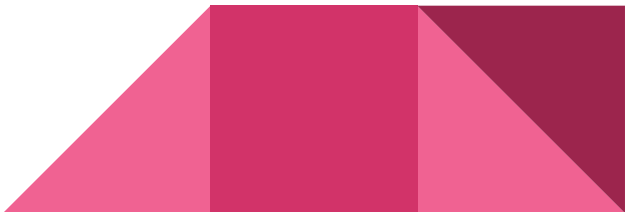
Only reference the actor's non-verbal cues such as body language, facial expressions, vocal tones, etc. Do not make references to what the judge or attorney said, only non-verbal cues accompanying or reacting to utterances.

Examples:

"The judge leaned forward."

"The judge looked at her notes."

"The attorney spoke loudly."



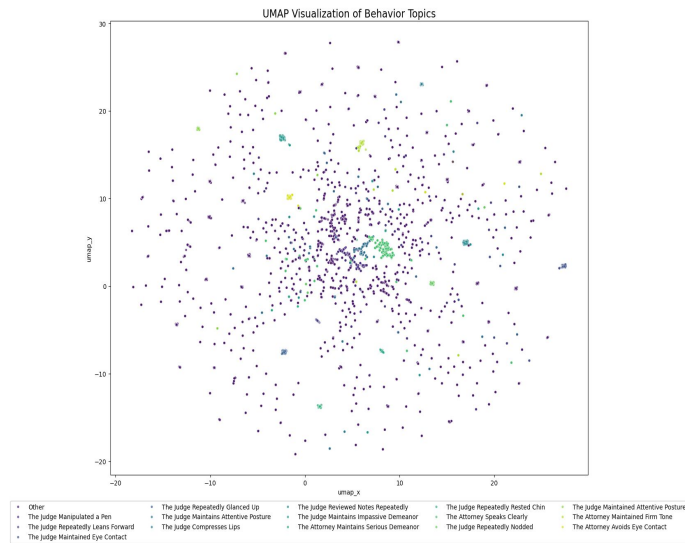
Examples of Extracted Behavior

19-50345 USA v Ernesto Medina

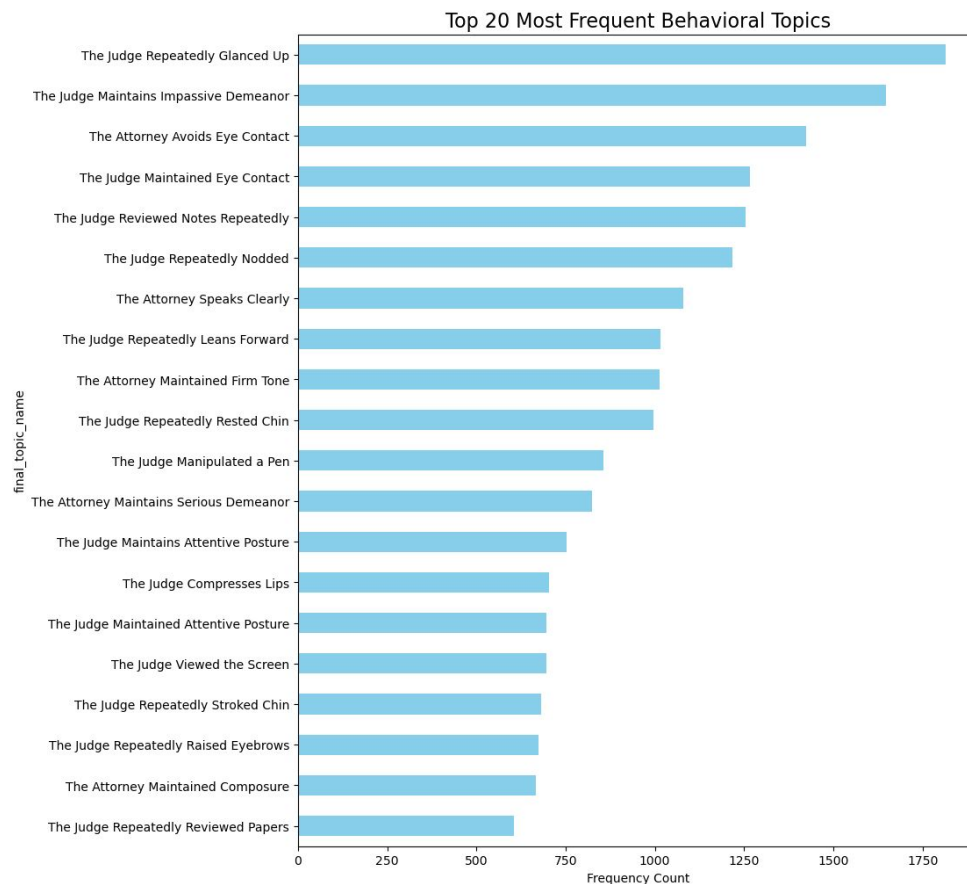
Timestamp	Behavior Description	Clip
00:00:00 00:00:06 00:00:09 00:00:11 00:00:16	The judge was looking down at his tablet. The judge looked up from his tablet toward the attorney. The judge picked up a white tissue. The judge leaned forward in his chair. The judge wiped his nose with the tissue. (Judge Watford)	 The clip consists of two stacked video frames showing the interior of a courtroom. The text overlay on both frames reads: "RICHARD H. CHAMBERS U.S. COURTHOUSE COURTROOM 3 PASADENA, CALIFORNIA". The frames show a judge sitting at a bench and an attorney sitting at a table.
00:12:32 00:12:48	The attorney gestured with an open palm. The attorney paused to listen to the judge's question. (Attorney)	 The clip consists of two stacked video frames showing the interior of a courtroom. The text overlay on both frames reads: "RICHARD H. CHAMBERS U.S. COURTHOUSE COURTROOM 3 PASADENA, CALIFORNIA". The frames show an attorney sitting at a table.

Topic Modeling Procedure

1. Encode each statement with BERT-based Sentence Encoder
 - a. **BERT**: Bi-directional Encoder-Decoder Transformer. Building block for LLM-based models
2. Dimension reduction with UMAP algorithm, cluster with HDBSCAN
 - a. **UMAP**: Uniform Manifold Approximation and Projection. Technique for lowering dimensionality (number of columns) in a dataset
 - b. **HDBSCAN**: Hierarchical Density-Based Spectral Clustering of Applications with Noise. Similar to k-means, clusters data into groups (across rows)
3. Topic model using tf-idf
 - a. **Term Frequency-Inverse Document Frequency**: Upweight words the more times they appear within a doc, downweight the more they appear across docs
 - b. Cluster words that co-occur together frequently
4. Label each topic with Gemini-2.5-pro using same [actor][verb][noun] structure
5. Iterate and merge similar topics with gemini until we reach ~100 distinct topics



Example Final Topics



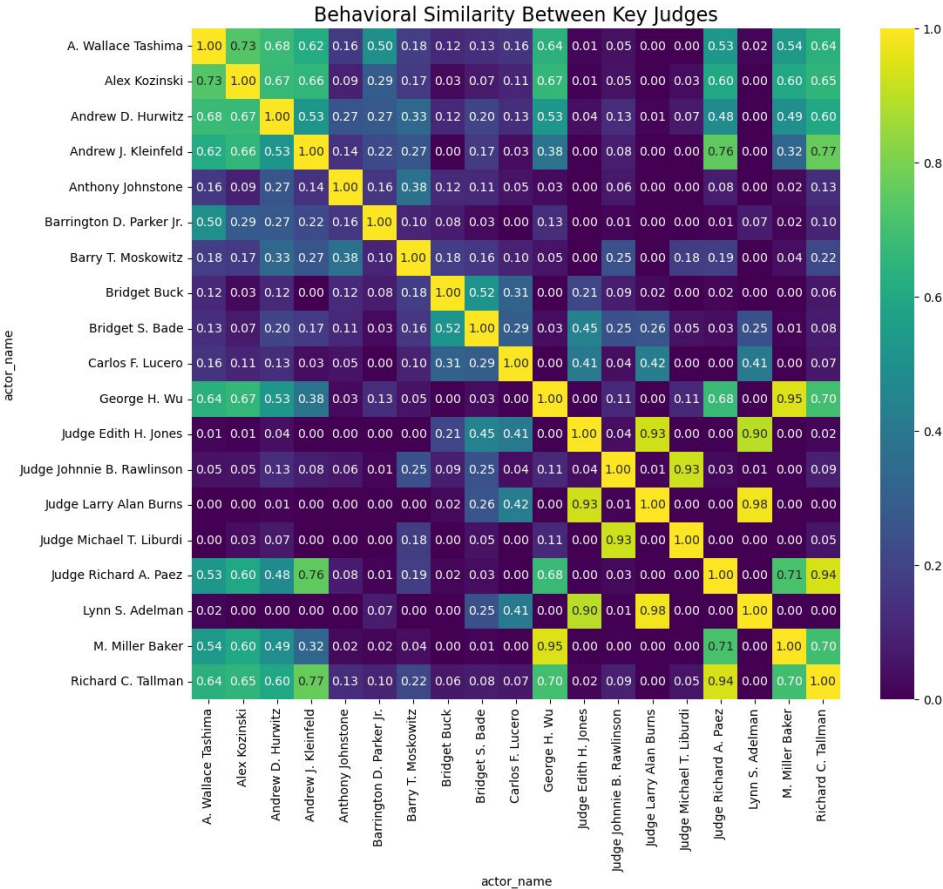
OLS on Topics

$$TopicShare_{ik} = \beta_0 + \beta_1 Female_i + \beta_2 Republican_i + \beta_3 Age_i + \beta_4 Modality_i + \epsilon_{ik}$$

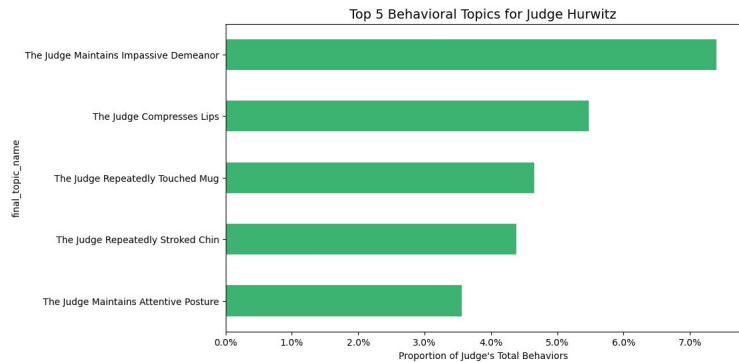
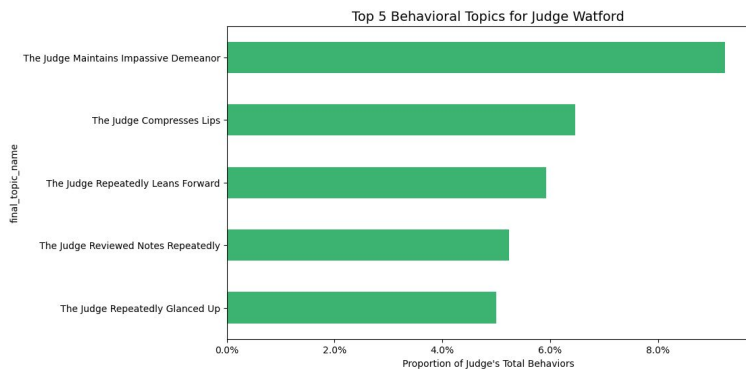
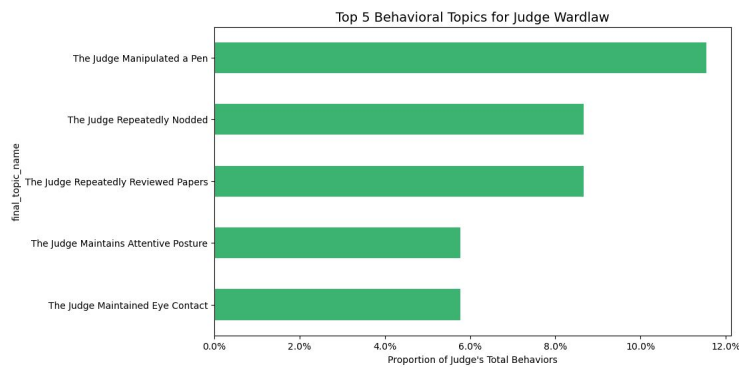
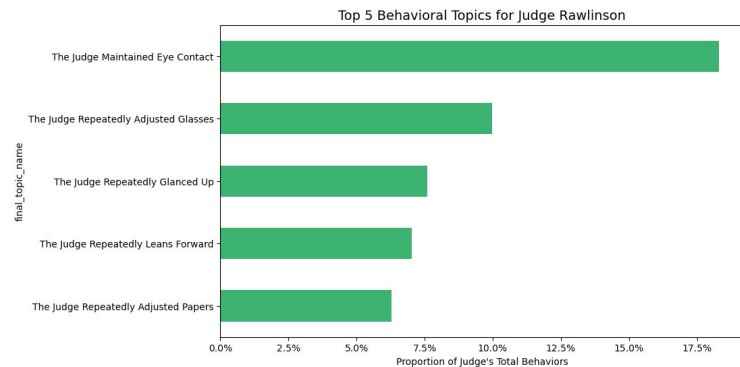
- Series of OLS regressions on each topic
- Share for judge i and topic k across judge characteristics and modality



Judge Vector Similarity

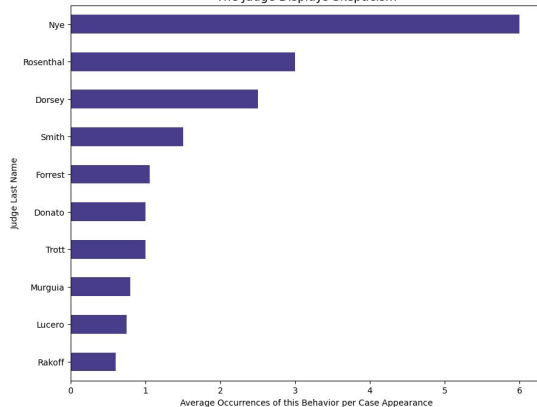


Judge Behaviors

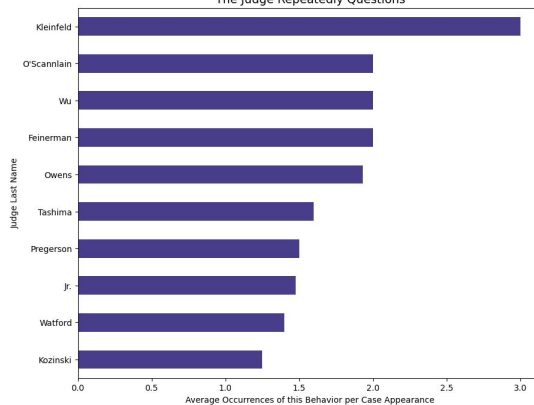


Ranking Judges within Topics

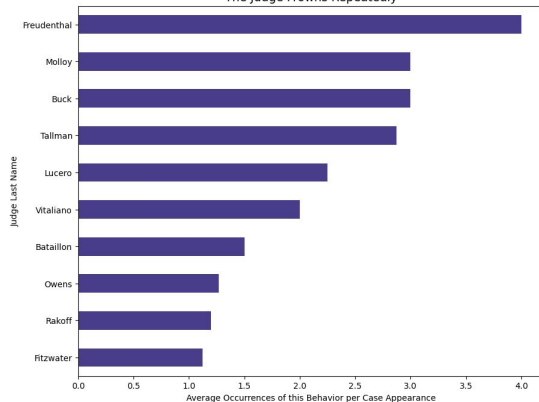
Top 10 Judges by Frequency of Behavior (Normalized by Case Count):
'The Judge Displays Skepticism'



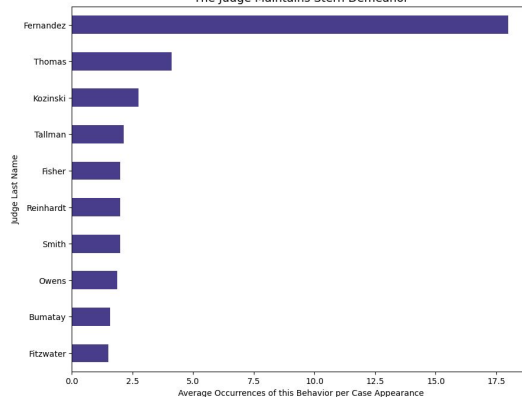
Top 10 Judges by Frequency of Behavior (Normalized by Case Count):
'The Judge Repeatedly Questions'



Top 10 Judges by Frequency of Behavior (Normalized by Case Count):
'The Judge Frowns Repeatedly'



Top 10 Judges by Frequency of Behavior (Normalized by Case Count):
'The Judge Maintains Stern Demeanor'





Empirical
Application: Who
Receives More
Questions?

Does the Losing Side Receive More Questions?

The New York Times

When the Justices Ask Questions, Be Prepared to Lose the Case

Share full article



By **Adam Liptak**

May 25, 2009

WASHINGTON

A few years ago, a second-year law student at Georgetown unlocked the secret to predicting which side would win a case in the Supreme Court based on how the argument went. Her theory has been tested and endorsed by Chief Justice John G. Roberts Jr., and has been confirmed by elaborate studies from teams of professors.

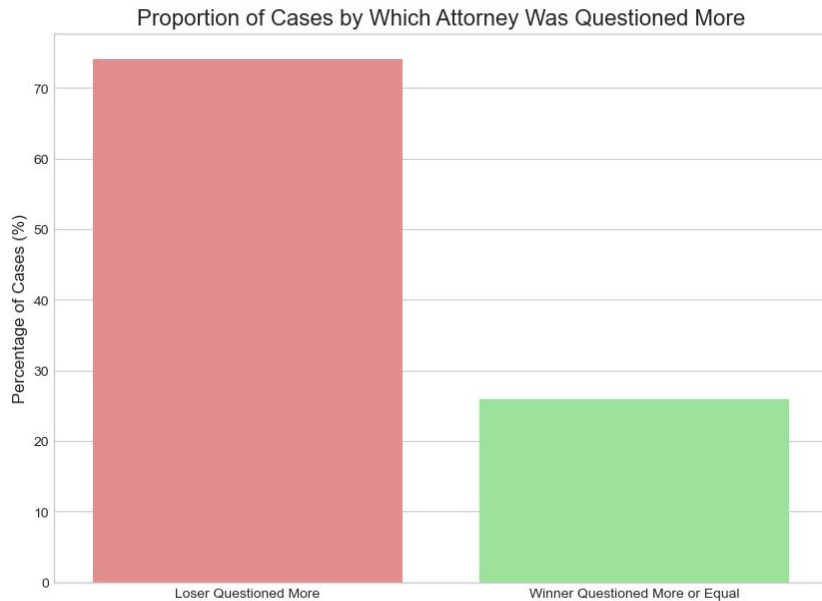
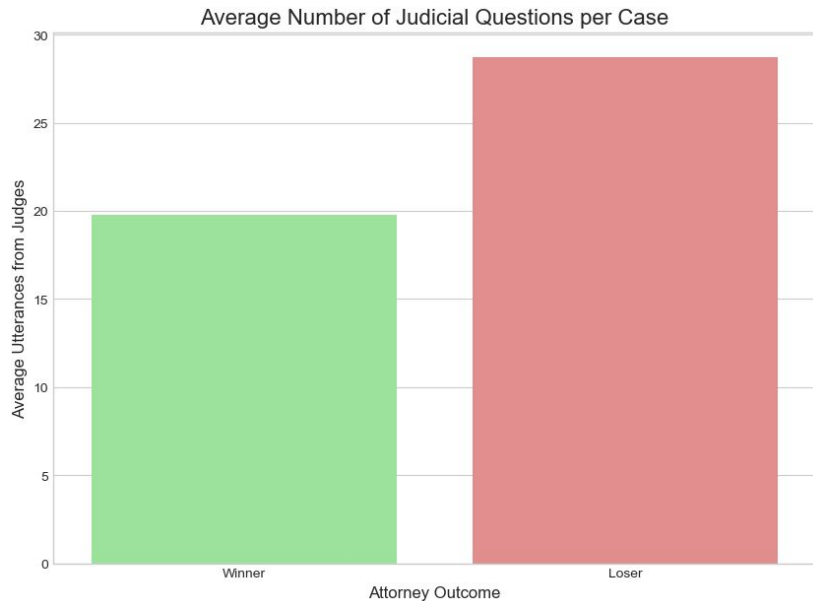
“The bottom line, as simple as it sounds,” said the student, Sarah Levien Shullman, who is now a litigation associate at a law firm in Florida, “is that the party that gets the most questions is likely to lose.”

Chief Justice Roberts heard about Ms. Shullman’s [study](#) while he was a federal appeals court judge, and he decided to test its conclusion for himself. So he picked 14 cases each from the terms that started in October 1980 and October 2003, and he started counting.

“The most-asked-question ‘rule’ predicted the winner — or more accurately, the loser — in 24 of those 28 cases, an 86 percent prediction rate,” he told the Supreme Court Historical Society in 2004.



More Questions Are Directed at the Loser in Aggregate and By Case



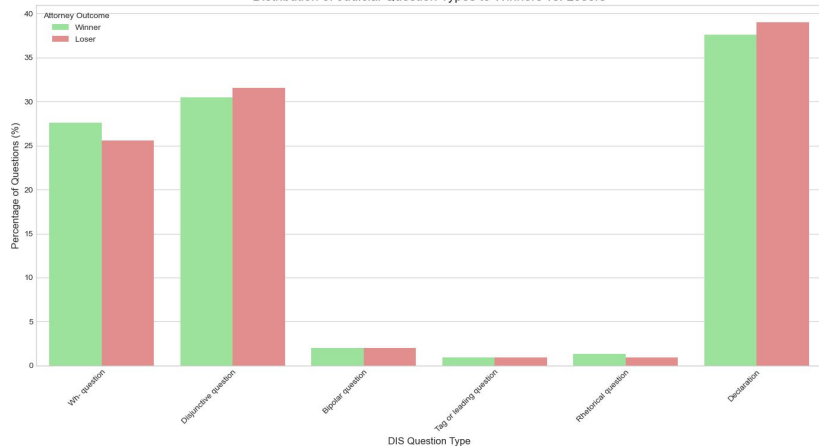
Phillips and Carter Typology of Information-Seeking

Table 1: Information-Seeking Typology for Oral Argument Utterances (Adapted from Phillips & Carter 2010)

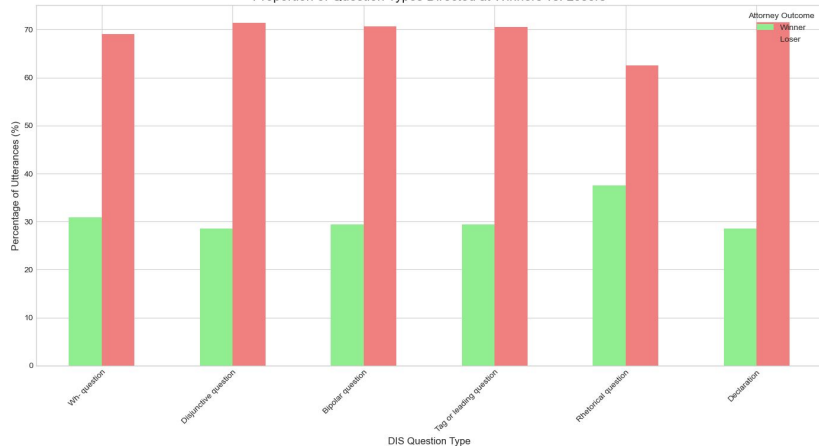
Level	Type	Explanation / Example
6	Declaration	Statement without a question mark. <i>Example:</i> “The First Amendment does not apply to this case.”
5	Rhetorical question	A question to which no answer is sought. <i>Example:</i> “How in the world am I supposed to believe your claim?”
4	Tag or leading question	A leading question, often framed negatively. <i>Example:</i> “You are claiming the First Amendment applies, are you not?”
3	Bipolar question	A question with only two answer options (yes/no, true/false). <i>Example:</i> “Are you claiming the First Amendment applies to this case?”
2	Disjunctive question	Closed question with an open-ended add-on. <i>Example:</i> “Are you claiming the First Amendment applies, or are you claiming something else?”
1	Wh- question	Open-ended question using who, what, when, where, why, or how. <i>Example:</i> “In what way are you claiming the First Amendment applies to this case?”

Losers Get More Questions in Disjunctive and Declaration Types

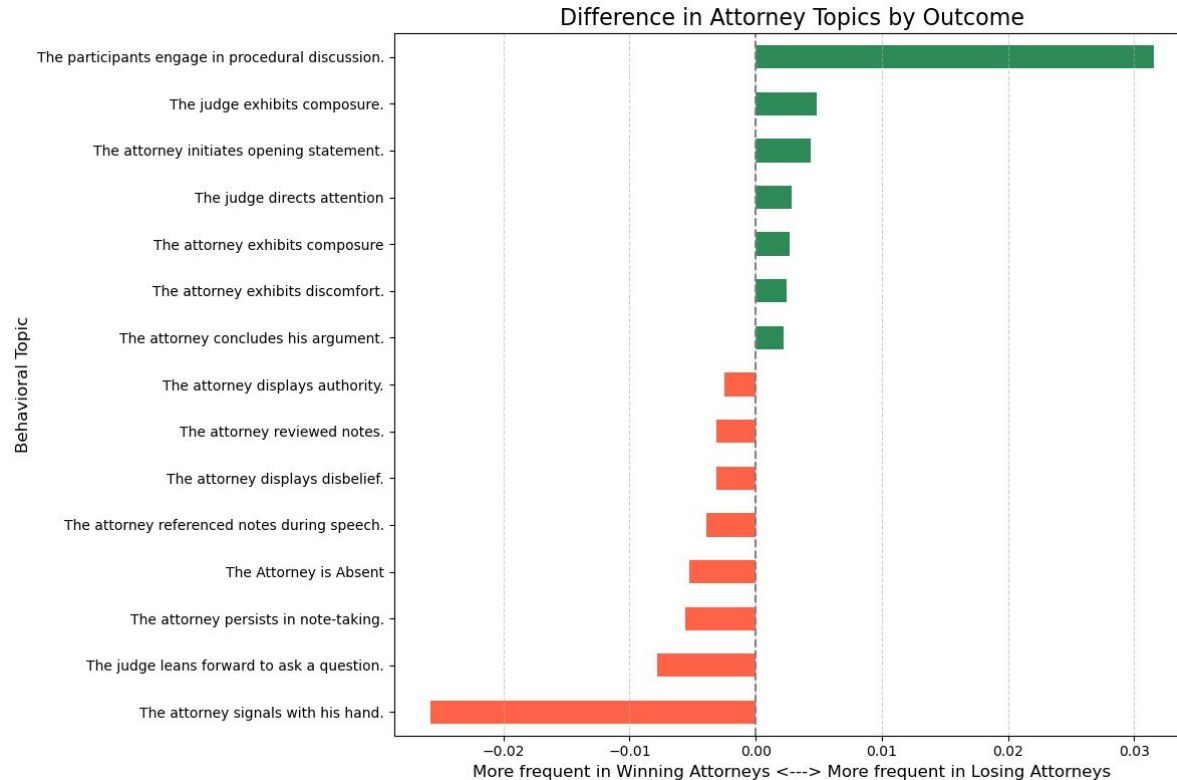
Distribution of Judicial Question Types to Winners vs. Losers



Proportion of Question Types Directed at Winners vs. Losers



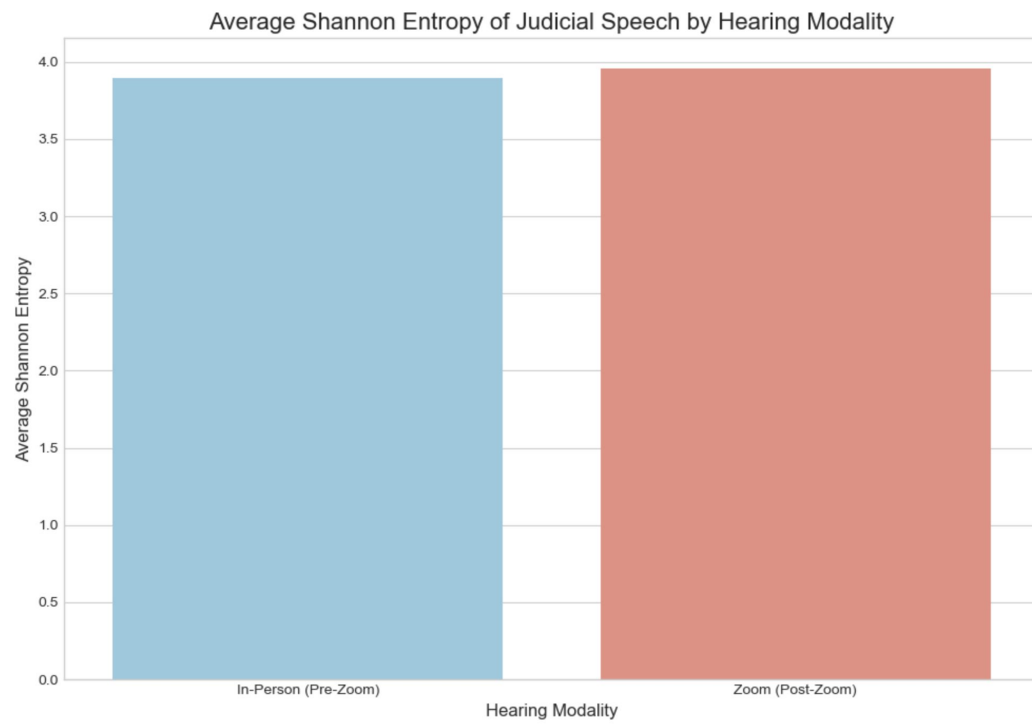
Topics Across Winning and Losing Attorneys



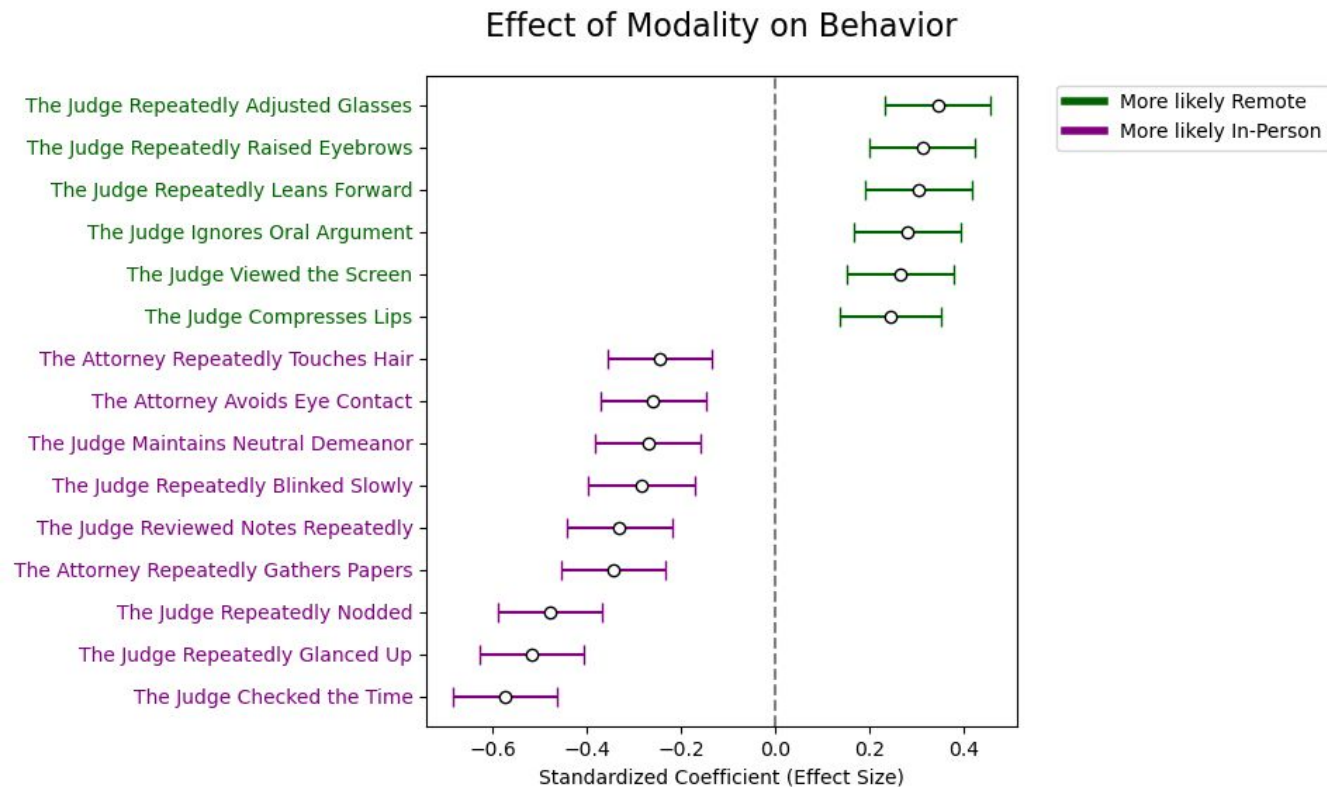


Empirical Application: Cameras in the Courtroom

Complexity of Judicial Speech Does Not Differ In-Person or Over Zoom



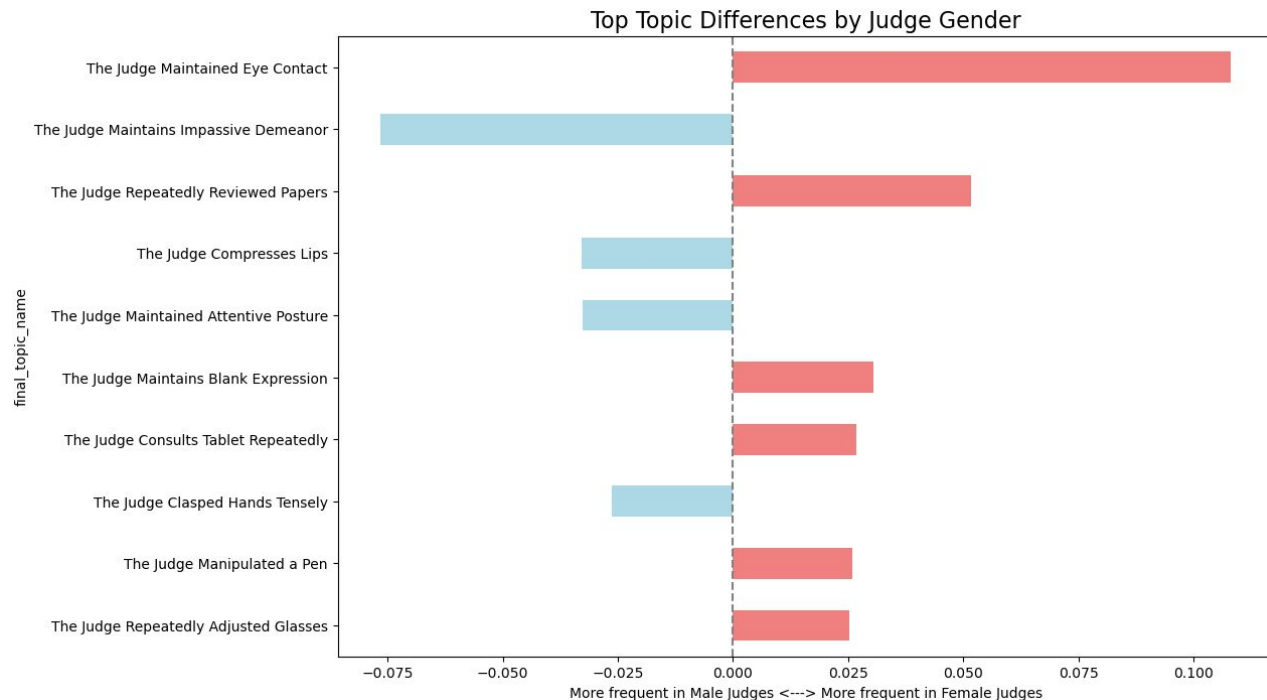
Differences in Behavior Across Modalities





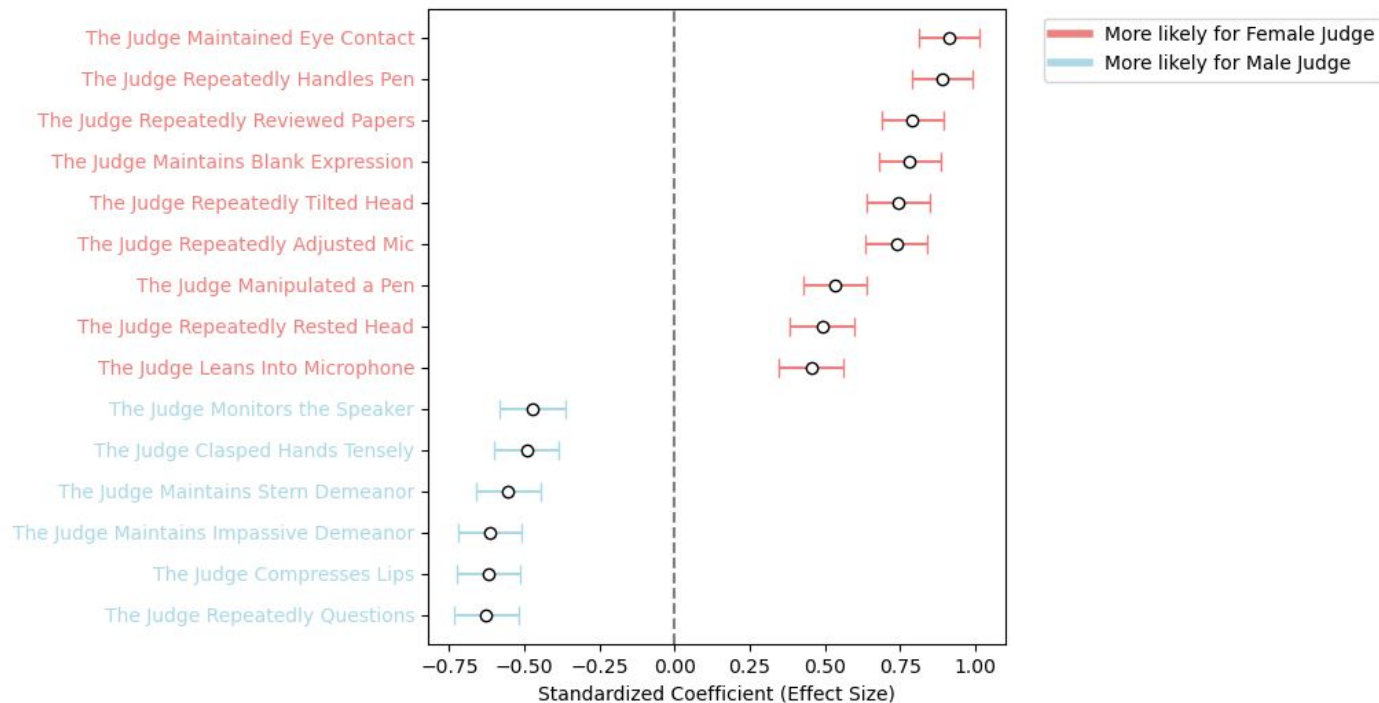
Empirical Application: How do Judges React to Gender?

Descriptive Results: Topics Across Male and Female Judges



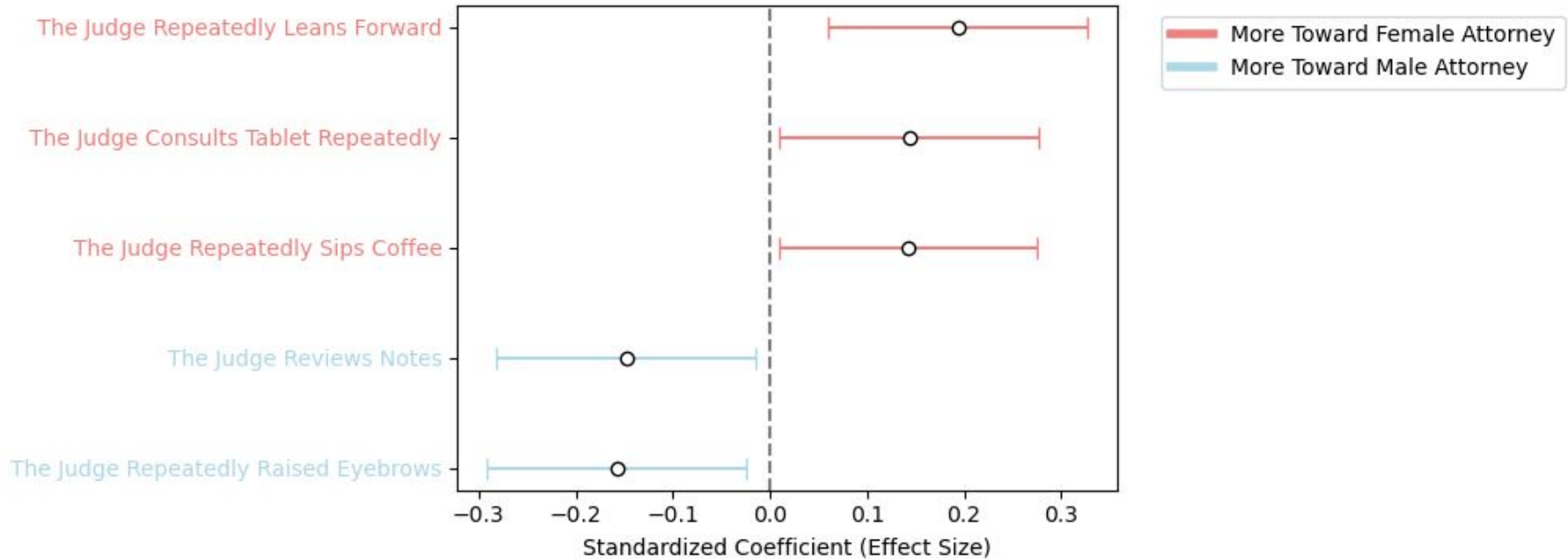
Regression Analysis: Topics by Gender

Effect of Judge Gender on Judge Behavior



Regression Analysis: Judge Reactions to Attorney Gender

Effect of Attorney Gender on Judge Behavior

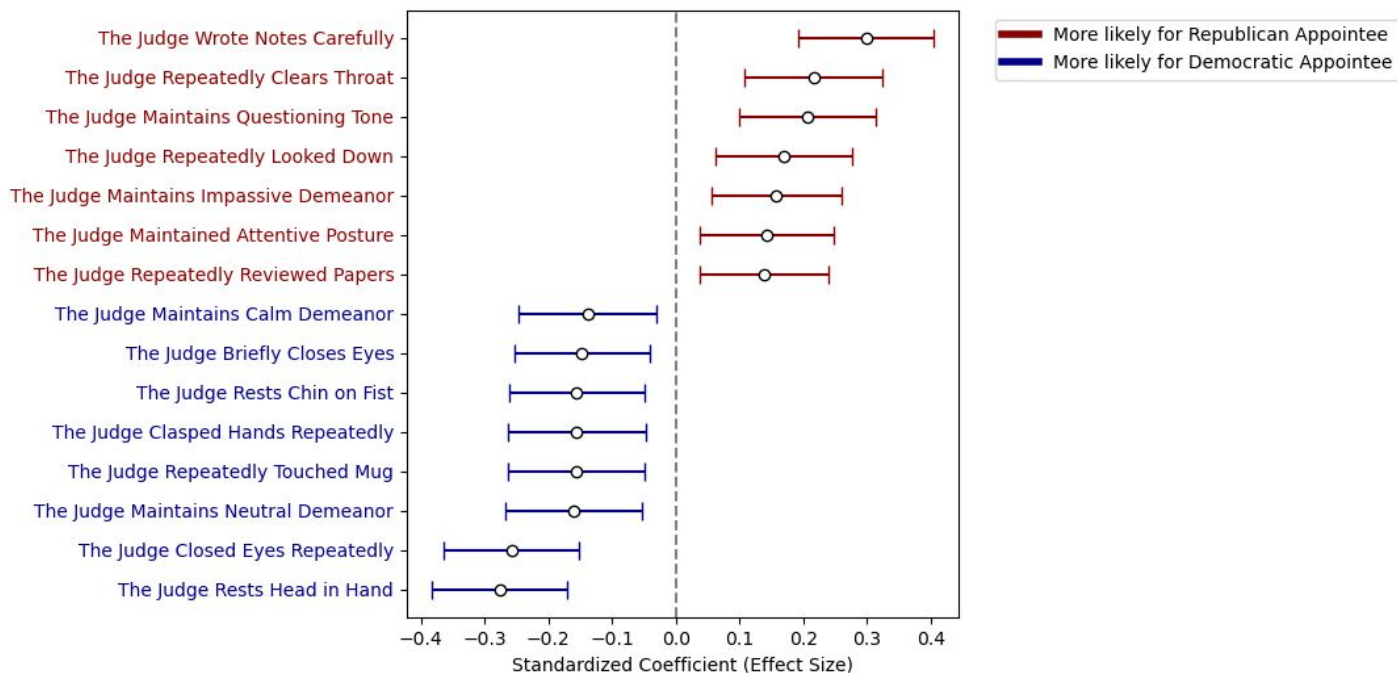




Empirical Application: Do Behaviors Differ by Party?

Regression Analysis: Topics by Party of Appointing President

Effect of Appointing Party on Judge Behavior



Conclusion

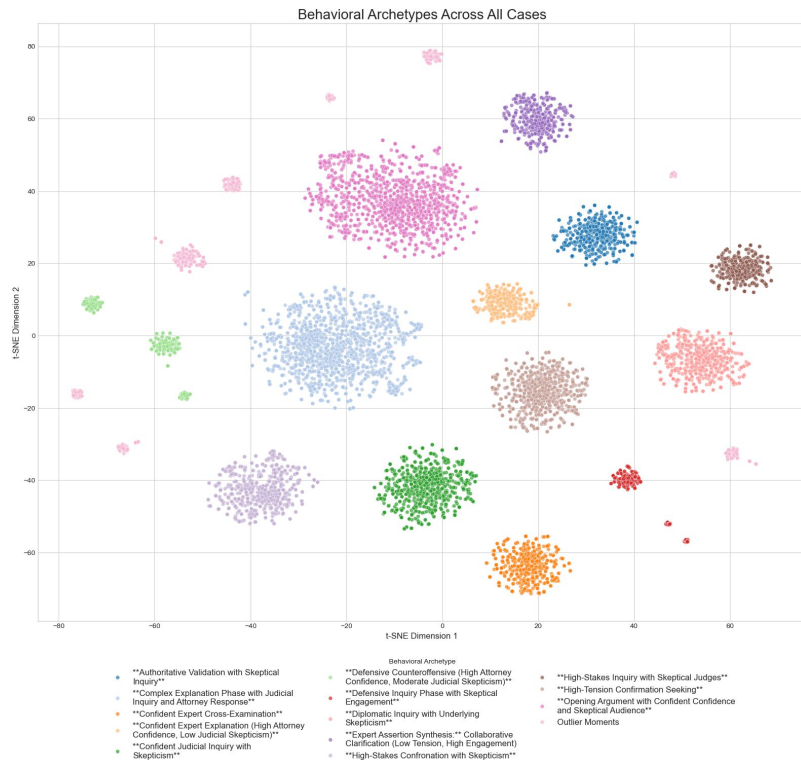


Concluding Thoughts

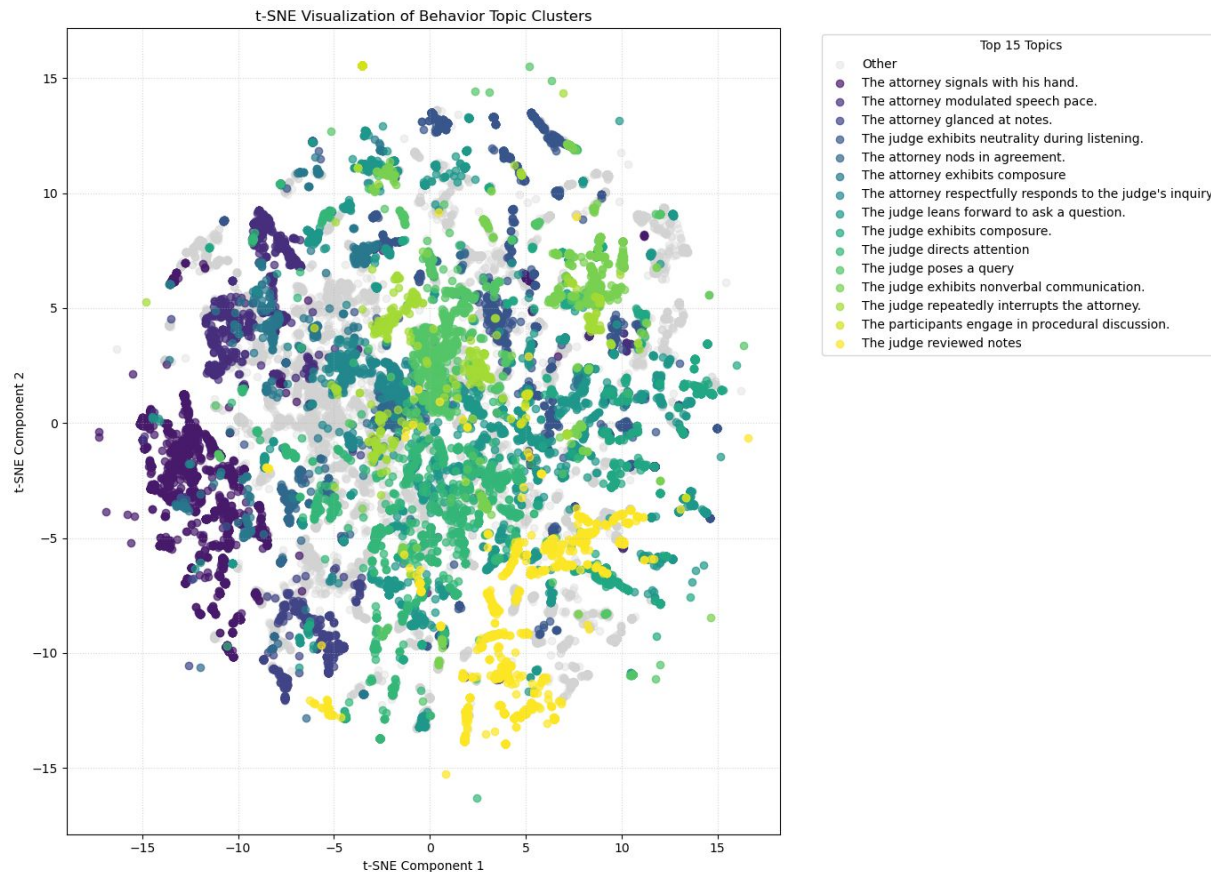
- Oral arguments are crucial for information gathering and AI can scale previous theories that focused on SCOTUS.
- Computational methods can be used to create narratives that are amenable to quantitative analysis
- Multimodal AI opens up empirical legal studies to new types of data

Extra Slides

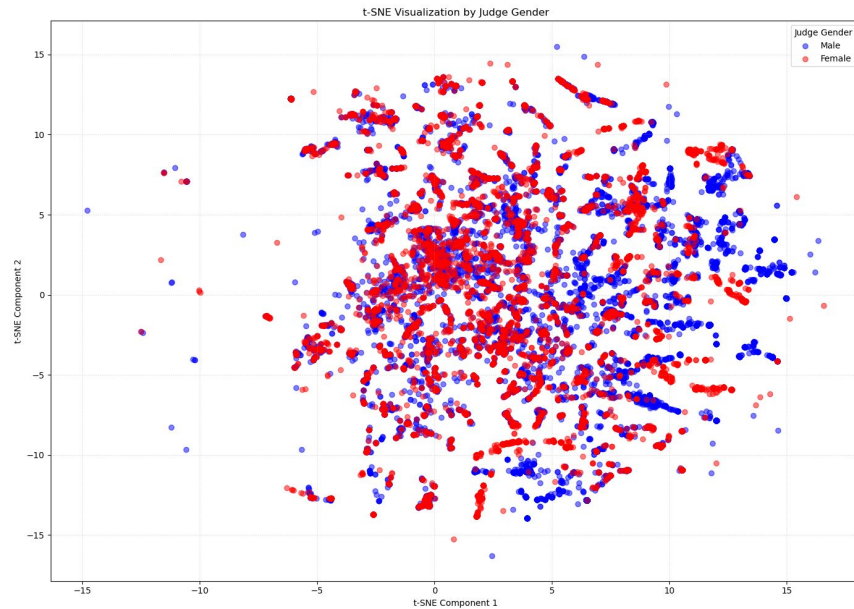
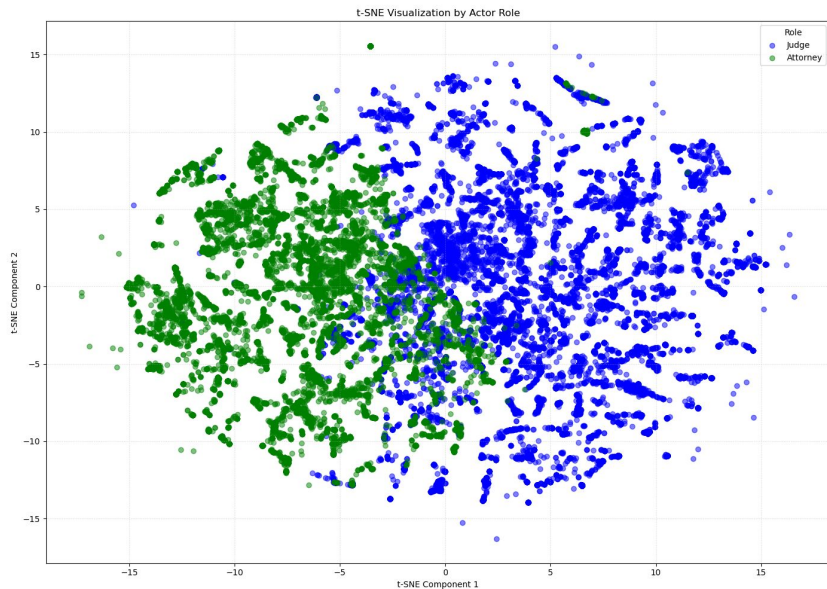
Behavioral Clusters from UMAP and HDBSCAN on Structured Data



Topic Clusters with t-SNE on Initial BERT Topics



t-SNE by Role and Gender



AI Summary of Structured Data

Case 01-71934 PUC of State of CA v. FERC

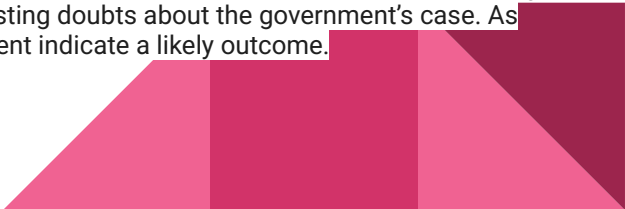
The hearing opens with the panel of judges displaying attentive yet reserved body language as the attorney for the petitioner presents their argument. Judge McKeown leans forward, pen poised, signaling engagement with the petitioner's points, while Judge Rawlinson's gaze shifts frequently between the speaker and her notes, indicating close tracking of the details. During a critical exchange on regulatory authority, Judge Smith's slight frown and head tilt reflect skepticism about the petitioner's interpretation. As the respondent rises, the mood in the room shifts: judges become visibly more animated, interrupting with pointed questions and occasionally exchanging glances. Judge McKeown's tone sharpens, and Rawlinson nods in agreement with several of the respondent's clarifications. By the close of the argument, the judges' body language is notably more open toward the respondent, with relaxed postures and supportive facial expressions, suggesting receptivity.

Winner prediction: respondent

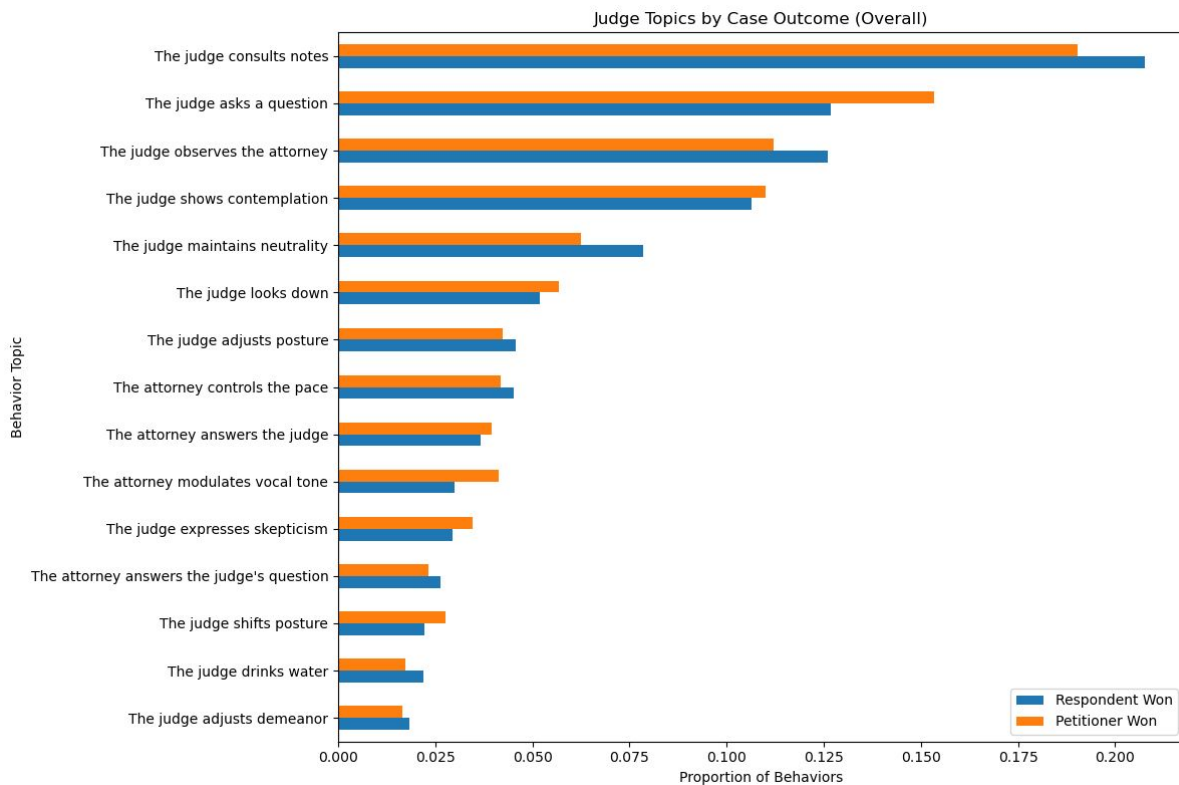
Case 06-75149 Lopez-Gomez v. Jefferson Sessions

From the outset, the panel appears sympathetic to the petitioner, as indicated by Judge Berzon's supportive nods and Judge Callahan's measured, encouraging tone. The judges offer the petitioner ample space to elaborate, often reframing questions in a way that helps clarify legal points. When the government's attorney begins, the atmosphere grows more tense. Judge Berzon leans back and folds her arms, while Judge Callahan's expression hardens, signaling skepticism. The panel presses the government attorney with challenging hypotheticals and critical follow-up questions, rarely breaking eye contact. Throughout the respondent's segment, the judges' posture and tone remain defensive, suggesting doubts about the government's case. As the hearing concludes, the judges' engagement with the petitioner and visible frustration with the respondent indicate a likely outcome.

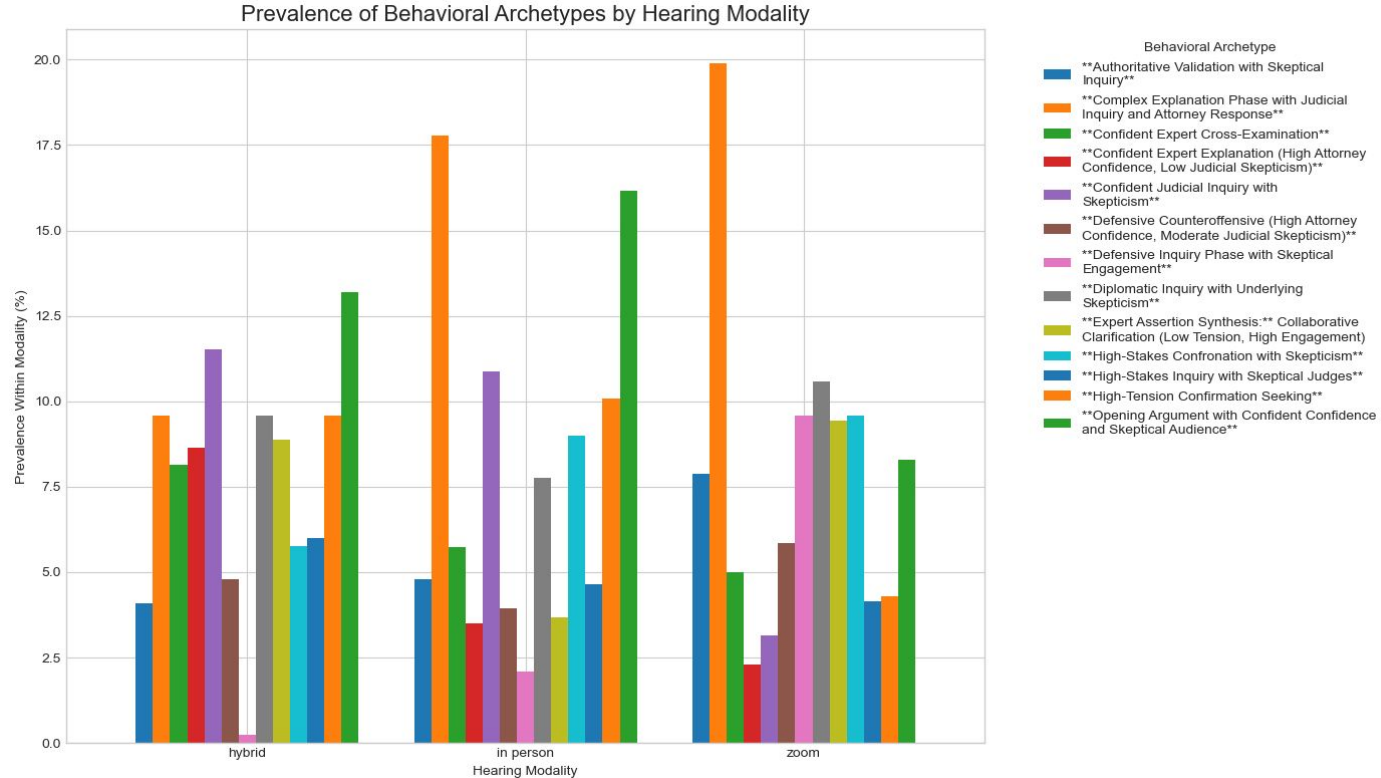
Winner prediction: petitioner



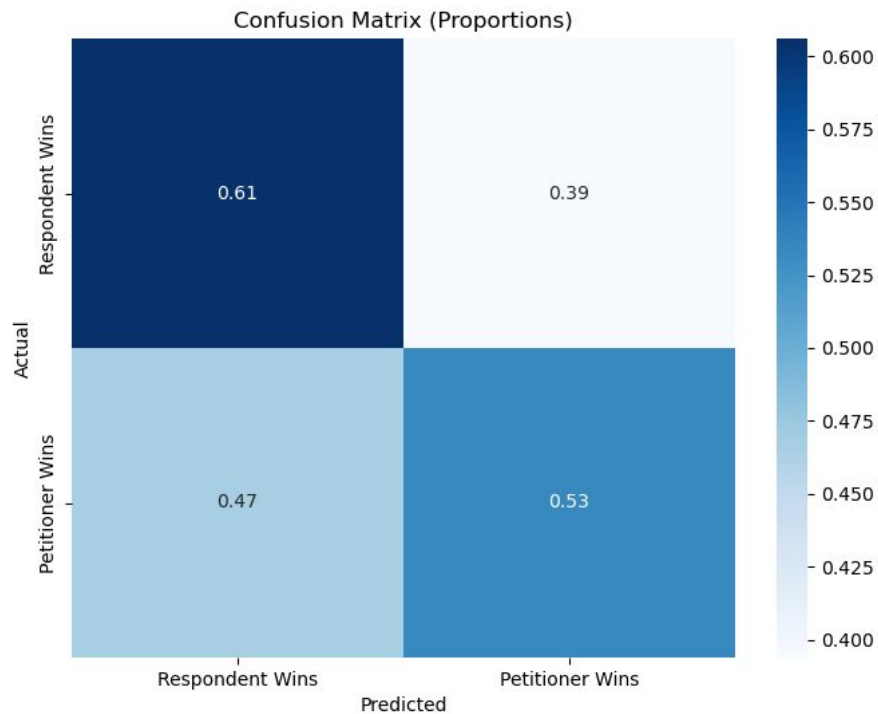
Judge Topics by Outcome



Behavioral Clusters by Modality



AI for Court Forecasting



Additional Gender Results

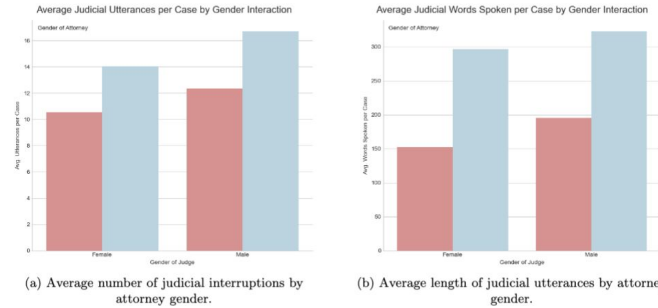


Figure 12: Judicial Interruptions by Attorney Gender. Panel (a) shows the average number of interruptions; Panel (b) shows average utterance length.

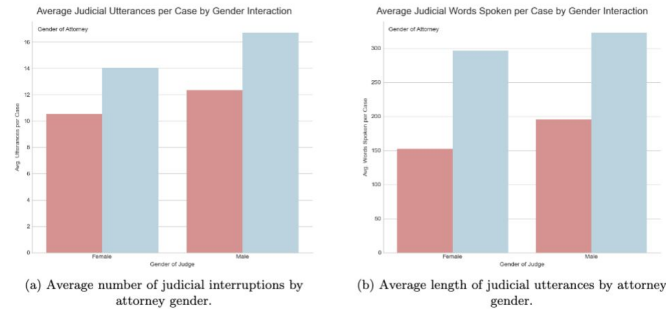
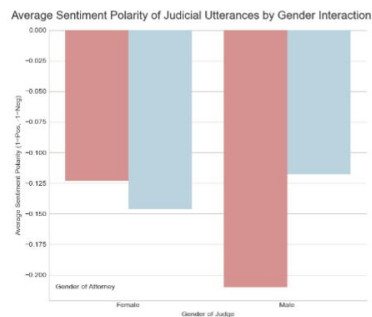
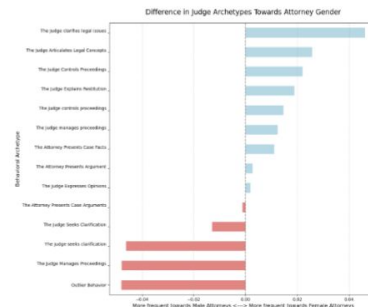


Figure 13: Judicial Interruptions by Attorney Gender. Panel (a) shows the average number of interruptions; Panel (b) shows average utterance length.

Additional Gender Results

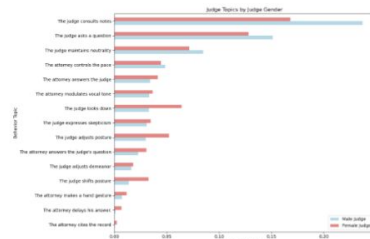


(a) Average sentiment of judicial speech directed at attorneys by gender.

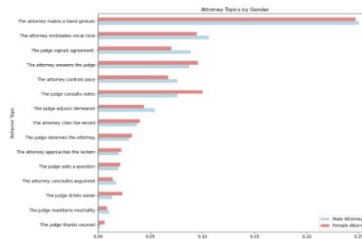


(b) Behavioral archetype distribution by role and gender.

Figure 14: Sentiment and Behavioral Archetypes by Gender. Panel (a) reports average sentiment; Panel (b) shows role-specific archetype distributions between behaviors directed at male and female attorneys.



(a) Judge Topics by Gender.



(b) Attorney Topics By Gender.

Figure 15: Analysis of Judge and Attorney Topics by Gender. Panel (a) shows the distribution of topics by judge gender. Panel (b) shows the distribution of topics by attorney gender

Additional Table - *Confident Judicial Inquiry with Skepticism*

Docket	Case	Timestamp	Excerpt	Clip
13-10397	USA v. Jose Medina-Carrasco	[02:46:85]	"Except if we were to accept your argument, then a waiver would never be valid if a good claim is lurking in the case as opposed to a worthless claim. And that would essentially do away with the waiver because in every case we'd be looking to see was there a valid claim and if so, then all bets are off and the waiver is not to be afforded any effect."	 JAMES R. BROWNING U.S. COURTHOUSE COURTROOM I SAN FRANCISCO, CALIFORNIA
17-73053	Hao Lin v. William Barr	[03:00:99]	"Well, there's a presumption that arises if he demonstrates past persecution, there's a presumption of future persecution which the government has an opportunity to rebut."	 RICHARD H. CHAMBERS U.S. COURTHOUSE COURTROOM I PASADENA, CALIFORNIA
15-73638	Patrick Thiongo v. Jefferson Sessions	05:25:47]	"But the health—remind me—the health issues, I remember, like, I don't know if it was high blood pressure or something less disabling, but not so severe that you couldn't bring yourself to file an application. It seemed to me that probably everyone who eventually makes their way to the United States claiming asylum could claim some adverse health effects. And if so, I just wanted to hear more from you about that."	 WILLIAM K. NAKAMURA U.S. COURTHOUSE 7TH FLOOR COURTROOM, COURTROOM 2 SEATTLE, WASHINGTON

Additional Table - Defensive Counteroffensive (High Attorney Confidence, Moderate Judicial Skepticism)

Docket	Case	Timestamp	Excerpt	Clip
16-10017	USA v. Lidia Rodriguez	[03:27]	"Judge Friedland: Yeah, that's the point. So the instruction that the jury convicted under did talk about knowledge of the facts, and I think if you have knowledge of these facts, I don't know how anyone doesn't draw the inference. I mean, it seems like at that point you need to rely on the idea that you would have knowledge that this isn't his ID and yet not worry that he is not trying to sneak in the country without valid documentation. And that inference seems so obvious that I don't know how there's not a harmless error here.	
16-10017	USA v. Lidia Rodriguez	[11:50]	"The district court did not consider the fact that they had an address on the card, and that they didn't give it to the defense attorney or try to contact. That's something that just was not discussed. And it wasn't the defendant's burden to bring that up. It was the government's burden to demonstrate unavailability."	
15-50508	USA v. Jorge Nunez-Duenas	[12:30:84]	"Two brief points, your honor. First, with respect to Marin-Cuevas, in addition to the points Judge Berzon made about the distinctions between this case and that case, I think an important fact in that case was that the underlying documents were fingerprint matched. So there was a basis to judge the reliability. Here we don't know the underlying reliability of this database—whether it is even a database or what the references are.	